Marmara Lectures

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Differentiation and Integration of functions

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Differentiation and integration of functions.

Theorem: Derivative of unit step function is dirac delta

$$\frac{d}{dt}U(t) = \delta(t)$$

Proof: Applying the definition of derivative

$$\frac{d}{dt}U(t) = \lim_{\varepsilon\to 0}\frac{U(t+\varepsilon)-U(t-\varepsilon)}{2\varepsilon} = \lim_{\varepsilon\to 0}\frac{\chi_{[-\varepsilon,\varepsilon]}}{2\varepsilon} = \delta(t)$$

More generally, differentiation of any discontinuity gives a dirac delta.

Theorem: Integration of dirac delta is unit step.
Proof:

$$\int_{-\infty}^{t}\delta(s)ds = \begin{cases}0, & t\leq 0\\ 1, & 0<t\end{cases}$$

or,

$$\int_{-\infty}^{t}\delta(s)ds = U(t)$$

Example: Consider the function $f(t)$. at $t=0$ this function is continuous and its derivative is also continuous.

$$f(t) = (at^3 + bt^4 + ct^5 + dt^6)U(t)$$

at $t = 0$ this function is continuous and its derivative $Df(t)$. is also continuous:

$$\begin{aligned}\frac{d}{dt}\left[(at^3 + bt^4 + ct^5 + dt^6)U(t)\right] &= \frac{d}{dt}\left(at^3 + bt^4 + ct^5 + dt^6\right)U(t) + \left(at^3 + bt^4 + \right. \\ &= (3at^2 + 4bt^3 + 5ct^4 + 6dt^5)U(t) + \left(at^3 + bt^4 + \right. \\ &= (3at^2 + 4bt^3 + 5ct^4 + 6dt^5)U(t)\end{aligned}$$

What about the second derivative? $D^2 f(t) = (6at + 12bt^2 + 20ct^3 + 30dt^4)U(t)$ is continuous $t = 0$.

What about the third derivative? $D^3 f(t) = (6a + 24bt + 60ct^2 + 120dt^3)U(t)$. .Therefore the third derivative is not continuous at $t = 0$. It has a discontinuity of size $6a$.

Fourth derivative is $D^4 f(t) = 6a\delta(t) + (24b + 120ct + 3600dt^2)U(t)$. It has a dirac delta and a discontinuity at $t = 0$.

If we take one more derivative what will happen? We will see in the next section.

Differentiation and integration of dirac deltas.

One may be surprised that a function as strange as dirac delta may have derivative. Contrary to the intiution, it has well defined derivatives of any order. We will denote n th order derivative of dirac delta as $\delta^n(t)$

$$\frac{d^n}{dt^n}\delta(t) = D^n\delta(t) = \delta^{(n)}(t)$$

We can use integration by parts to write

$$\int_{-\infty}^{\infty} \delta^{(n)}(t)f(t)dt = \delta^{(n-1)}(t)f(t)\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \delta^{(n-1)}(t)Df(t)dt$$

Imposing the boundary conditions

$$\delta^{(n-1)}(t)f(t)\Big|_{-\infty}^{\infty} = 0$$

will result in

$$\int_{-\infty}^{\infty} \delta^{(n)}(t) f(t) dt = -\int_{-\infty}^{\infty} \delta^{(n-1)}(t) Df(t) dt$$

As $f(t)$ is completely general, the integrands must be equal

$$\delta^{(n)}(t) f(t) = -\delta^{(n-1)}(t) Df(t)$$

by iteration, we can get

$$\delta^{(n)}(t) f(t) = (-1)^n \delta(t) D^n f(t)$$

Summarizing our results:

$$\boxed{\delta^{(n)}(t) = -\delta^{(n-1)}(t) D = (-1)^n \delta(t) D^n}$$

which is a generalization of theProperty 6, Multiplication of dirac deltas with functions, defined in the previous chapter.

Example: Find $(3t^2 + 4t + 7)\delta(t)$
Answer: $7\delta(t)$

Example: Find $(3t^2 + 4t + 7)\delta^{(2)}(t)$
Answer: $6\delta(t)$

Example: Find $(3t^2 + 4t + 7)\delta^{(1)}(t)$
Answer: $-4\delta(t)$

Example: Consider the function $f(t)$

$$f(t) = \begin{cases} 0, & t \leq 0 \\ a + bt + ct^2 + dt^3, & 0 < t \end{cases}$$

Or, equivalently, $f(t) = (a + bt + ct^2 + dt^3)U(t)$. Compute its derivatives. Note that there is a discontinuity at $t = 0$.
Answer:

$$\begin{aligned} Df(t) &= a\delta(t) + (b + 2ct + 3dt^2)U(t). \\ D^2 f(t) &= a\delta^{(1)}(t) + b\delta(t) + (2c + 6dt)U(t). \\ D^3 f(t) &= a\delta^{(2)}(t) + b\delta^{(1)}(t) + 2c\delta(t) + 6dU(t). \\ D^4 f(t) &= a\delta^{(3)}(t) + b\delta^{(2)}(t) + 2c\delta^{(1)}(t) + 6d\delta(t) \end{aligned}$$

Exercise: Integrate $D^4 f(t)$ found above four times to get back $f(t)$.
Answer: Left to the student.

Exercise: Compute the derivatives of the function $g(t) = (4e^{-2t} + 3e^{-3t} + e^t)U(t)$
.
Answer:

$$
\begin{aligned}
Dg(t) &= 8\delta(t) + (-8e^{-2t} - 9e^{-3t} + e^t)U(t).\\
D^2g(t) &= 8\delta^{(1)}(t) - 16\delta(t) + (16e^{-2t} + 27e^{-3t} + e^t)U(t).\\
D^3g(t) &= 8\delta^{(2)}(t) - 16\delta^{(1)}(t) + 44\delta(t) + (-32e^{-2t} - 84e^{-3t} + e^t)U(t).\\
D^4g(t) &= 8\delta^{(3)}(t) - 16\delta^{(2)}(t) + 44\delta^{(1)}(t) - 115\delta(t) + (64e^{-2t} + 252e^{-3t} + e^t)U(t)
\end{aligned}
$$

Exercise: Integrate $Dg(t)$ to get back $g(t)$
Answer:

$$\int_{s=-\infty}^{t} Dg(s)ds = \int_{s=-\infty}^{t} (8\delta(s) + (-8e^{-2s} - 9e^{-3s} + e^s)U(s))ds$$

- for $t < 0$ this integral is 0
- for $t > 0$ this integral is
  $\int_{s=0^-}^{t} Dg(s)ds = 8 + (4e^{-2t} + 3e^{-3t} + e^t - 8) = 4e^{-2t} + 3e^{-3t} + e^t$

Joining the two parts we get $g(t) = (4e^{-2t} + 3e^{-3t} + e^t)U(t)$.

General rule for differentiation at discontinuities

Consider the function

$$y(t) = g(t)U(t)$$

where $g(t)$ is continuous at $t = 0$ with finite and continuous derivatives, $\frac{d^k}{dt^k} g(t) = g^{(k)}(t)$, of all orders.

What is the derivatives of $y(t)$ at $t = 0$, of all orders? Obviously, left and right derivatives will differ. Left derivatives of any order is zero. So, we only need to find the right derivatives at $t = 0^+$.

$$
\begin{aligned}
y(t) &= g(t)U(t)\\
Dy(t) &= g^{(1)}(t)U(t) + g(t)\delta(t) = g^{(1)}(t)U(t) + g(0)\delta(t)\\
D^2y(t) &= g^{(2)}(t)U(t) + g^{(1)}(t)\delta(t) + g(0)\delta^{(1)}(t) = g^{(2)}(t)U(t) + g^{(1)}(t)\delta(0) + g(0)\delta
\end{aligned}
$$

And the general rule is:

$$\boxed{D^k y(t) = g^{(k)}(t)U(t) + \sum_{i=0}^{k-1} g^{(i)}(0)\delta^{(k-1-i)}(t)}$$

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